Statistical Consulting Meeting Report

Date: Apr 1, 2026, 11:00 AM - 12:00 PM
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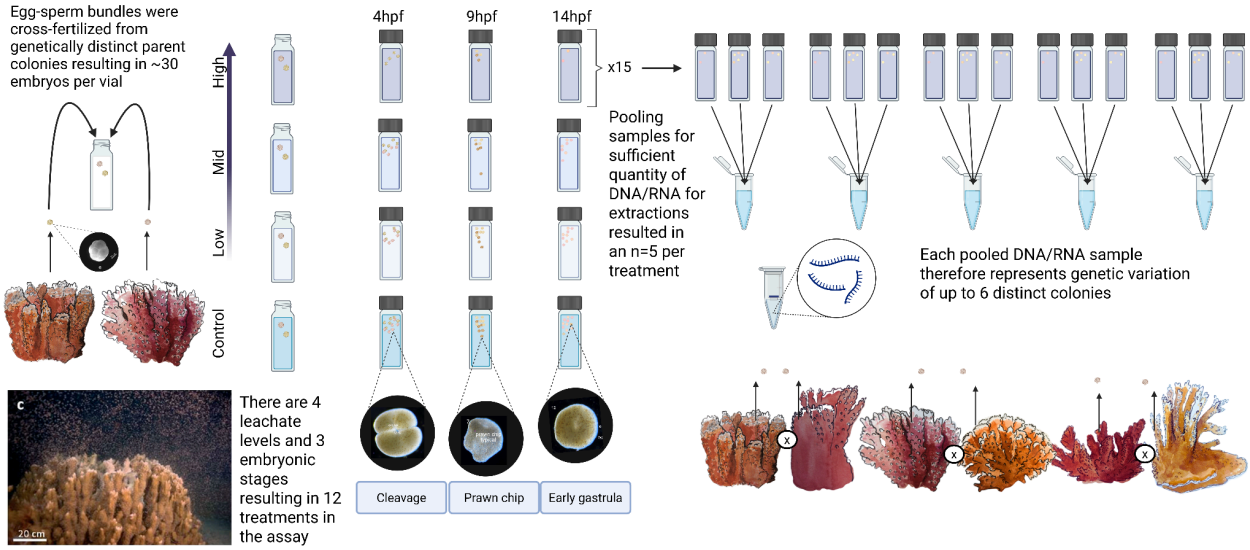
Project Goal and Research Questions

Background: The client's project investigates how different dosages of microplastic leachate, a chemical mixture that leaches from microplastics into seawater, affect embryonic development in the coral *Montipora capitata.* The client described her experimental design as follows: eggs released from coral specimens are dispersed into the water and fertilized at the water surface. Eggs are collected and added to 15 vials for each of the 4 leachate dosage conditions (control, low, mid, and high). The vials are then observed at three time points (4, 9, and 14 hours post fertilization, hpf), and embryos are categorized into 5 mutually exclusive development stages: egg, cleavage, morula, prawnchip, and early gastrula. The Figure below (see [Sarah's `coral-embryo-RNAseq` repository](#)) summarizes the data collection.



Due to the presence of unfertilized or dead embryos, whose counts are not tracked, the total number of counted embryos does not sum to a fixed total per vial. Because of that, modeling the development composition (i.e., the fraction of the total population in each development category) was deemed infeasible. The client's method of choice to analyze the data is a multivariate negative binomial generalized linear model (MNB-GLM) implemented via the *mvabund* package in R, which fits separate NB models to embryo counts in each development stage.

Project Goal: The goal of this project is to answer the question: "Does leachate affect the developmental timing and stage composition of developing coral embryos?" More specifically, the client is interested in assessing the adequacy of her proposed analysis based on the multivariate negative binomial generalized linear model (MNB-GLM) to answer this question.

Statistical Questions: relevant questions that were brought by the client and discussed during the appointment are:

1. Which claims about the effect of leachate are warranted under the current usage of the NB-GLM model?
2. How to formally test the claim that data are zero-inflated?
3. Why do aggregate tests (over all development categories) in *mvabund* show that the effect of treatment is significant, whereas the effect on each development stage is not?
4. How to integrate a model for the survival component and a model for the development component to make compositional claims about developmental timing defensible?

Recommended Action Plan

To answer these statistical questions, we recommend the following action plan:

Step 1: Discussing valid claims based on the current data analysis: our first observation is that the MNB-GLM model is an appropriate choice for multivariate count data of this kind. However, we would like to point out which claims are warranted under the current application in order to help the client discuss the limitations of her approach. The following are supported claims:

1. **Aggregate effect on the number of counts in each development category**: the significance of the interaction between treatment and time ("treatment:hpf") implies that the effect of leachate on the *counts of each development category* is time-dependent, meaning that counts differ between treatment groups across

time. Notice the emphasis on counts. In the current setup, it is not possible to claim that treatment shifts the proportion of embryos at a given stage, because the model does not condition on the total number of surviving embryos per vial. In this case, a reduction in counts for a given stage under high leachate could reflect slower development, higher mortality, or both.

2. **Absolute count claims**: beyond the analysis of the deviance table, the client can obtain level-specific information by running in R

   ```
   > summary(mod_all,  nBoot = 5000)
   ```

   Here, setting a large number of bootstrap replicates reduces the variability of the simulated p-values. This gives a coefficient table with separate rows for each of the treatment levels versus the control baseline. The client can then assess whether each dose differs from control over time. For instance, whether high leachate differs from control at 14 hpf for a specific stage. This gives more information than just whether treatment as a whole explains the deviance, as provided by the *anova* command.

Step 2: Formalizing the claim of zero-Inflation through a statistical test: the claim of zero-inflation can be formally tested by comparing the fit with the Negative Binomial GLM model and a zero-inflated negative binomial model through an analysis of deviance. Following our appointment, the client compared the fit of a standard NB model with a zero-inflated model using the *anova* function in *glmmTB.* The client concluded that zero-inflation is not necessarily supported by the data. We have reviewed the analysis and confirm that the approach is valid.

One important source of apparent zero excess is the presence of structural zeros. These are zeros that arise because a stage is biologically impossible at a given time point rather than from sampling variability. If structural zeros are present in the data, they may explain the apparent zero excess without requiring a zero-inflated model.

Step 3: Understand the driving factors behind significant aggregate results and null individual effects: in Wang et al., 2012, Fig.1, the authors of *mvabund* provide an analysis of deviance table that shows that treatment is deemed to have a significant effect on the abundance of Copepods through a multivariate test, but individual tests (i.e., species-by-species) are not deemed significant. The discussion in the Section "Testing Hypotheses about the Community-Environment Association" in Wang et al., 2012 can be relevant here. We believe that an important driving factor for this is that univariate p-values (i.e., a p-value for testing the significance of the interaction for each

development stage) are corrected for multiple comparisons, which can reduce power for detecting the effects.

Step 4: Modeling survival concomitantly to a development component to make compositional claims: In this step, we elaborate further on what can be done to ensure that compositional claims are backed by the model.

The client's current model is

```
> mod_all <- manyglm( Y ~ treatment * hpf,
        family = "negative.binomial",
        data = tidy_vials_stage_counts)
```

This model estimates the effect of treatment and time on the absolute count of embryos at each developmental stage. In this specification, the treatment coefficient reflects a mixture of survival and developmental effect, because leachate affects both how many embryos survive and how surviving embryos are distributed across developmental stages. We propose the following correction:

```
> tidy_vials_stage_counts <- tidy_vials_stage_counts |>
  mutate(N_obs = n_egg + n_cleavage + n_morula + n_prawnchip + n_earlygastrula)
> mod_offset <- manyglm( Y ~ treatment * hpf,
        family = "negative.binomial",
        offset = log(N_obs),
        data =tidy_vials_stage_counts)
```

This implementation uses an offset term. Here, *N_obs* is the total number of scored/classified embryos per vial. By including log(N_obs) as an offset, the model is modeling the expected count divided by N_obs, which is the fraction of scored embryos at each developmental stage. This ensures that treatment and time capture shifts in composition rather than shifts in absolute counts in each developmental stage.

Final Note

Sarah, you are welcome to return to consulting for further discussion if you think you need it. If you return during the Spring quarter 2026, please book into the same time slot, Wednesday at 11:00 AM, if possible, so that you meet with the same set of consultants. Additionally, please feel free to email me (mpotto@uw.edu) if you have any quick questions regarding this summary.